

Ernesto Rivera

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EDUCATION

Boston University

B.A. in Computer Science

Boston, MA

May 2026

- Coursework: Data Structures, Algorithms, Distributed Systems, Full-Stack, Artificial Intelligence, Machine Learning, Operating Systems, Software Engineering, Computational Probability, Logic Design, AI Ethics

EXPERIENCE

IBM

Software Developer

San Jose, CA

Jul. 2026 - Present

- Built **E2E validation and debugging tooling** from the ground up for a webhook-driven **AI-agent pipeline** in **watsonx Orchestrate**, enabling reliable QA intake and containment reviews.
- Selected to architect **testing infrastructure** spanning **watsonx Orchestrate**'s full-stack app and agentic workflows, backing a **30-person** team and leading **2 engineers**.

Wayfair

Software Engineer Intern

Boston, MA

Jun. 2025 - Aug. 2025

- Resolved a critical **26%** customer drop-off in the WFCC application funnel by owning the end-to-end deployment of a server-side **PHP** feature, delivering a state-aware banner that guided users through a key conversion point.
- Architected and implemented a scalable E2E test framework using **Playwright** and the **Page Object Model (POM)**, creating custom fixtures to solve complex state management via browser cookie injection.
- Prevented a high-risk deployment targeting a **14%** conversion gap by mapping complex user flows to **Mermaid** diagrams, ensuring safer production updates in collaboration with Product and Marketing stakeholders.

Global Scope Research Lab

Research Assistant

Boston, MA

Jan. 2024 - May 2024

- Analyzed a **32 GB**, 370 million-row dataset from the US Patent Directory to uncover innovation trends before and after Hurricane Katrina, identifying correlations between natural disasters and innovation.
- Collaborated with a Research Assistant using **NumPy** and **Dask** on a shared computing cluster, streamlining data processing to extract scalable insights into patent trends and support post-disaster policy analysis.

MyTA

Founding Engineer

Remote

Jul. 2023 - Mar. 2024

- Built an **OCR-to-Markdown** pipeline with **Meta AI's Nougat** for semantic retrieval of faculty materials.
- Indexed course content in **Pinecone** to power chatbot retrieval, supporting adoption across **9 universities**.

PROJECTS AND COMPETITIONS

Browser-Native Distributed LLM Training

Mar. 2026 - Aug. 2026

- Built a browser-native distributed GPT-2 trainer that partitions transformer layers across GPU-backed Chrome workers using **TensorFlow.js**, **WebRTC**, and **1F1B pipeline parallelism**.
- Validated a **774M-parameter GPT-2 Large model** across **6 Chrome workers in 3 AWS regions**, using **DiLoCo** and **SparseLoCo** to reduce synchronization traffic by approximately **65x**.
- Implemented **gradient checkpointing** and **8-bit Adam** in TensorFlow.js, reducing activation memory by approximately **60%** and optimizer-state memory by **4x**.

Is Your Agent Ready to Pay?

May 2026

- Built an **agentic commerce readiness scanner** that analyzes product identity, price certainty, payment routes, buyer-risk policies, and evidence quality to determine whether an **AI agent** can safely request wallet authorization.
- Engineered a commerce extraction and verification pipeline using **Node.js**, **Cheerio**, and **JSON-LD**, detecting checkout capabilities including **Stripe/Link**, digital wallets, UCP endpoints, and **x402** payment challenges.

Gloo Hackathon: *Best Technology - \$50,000 Prize*

Oct. 2023

- Pioneered the development of personalized faith-based AI counselors, accessible via targeted social media and search engine ads, which significantly enhanced outreach efficiency and achieved a **100x** cost reduction.
- Engineered an AI-driven **MetaHuman** backend with **Inworld AI** and **Unreal Engine**, enabling immersive **3D** interaction that improved user engagement, leveraging human-to-LLM retention stats.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, C, Java, Go, PHP, OCaml, SQL, GraphQL, Solidity

Frameworks/Libraries: LangGraph, LangChain, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Dask, Neo4j, Pinecone RAG, OCR (Nougat, MMOCR, DocTR), Next.js, React, Express.js, Django, Flask, FastAPI

Tools/Platforms: AWS, GCP, Azure, Docker, Kubernetes, Terraform, Kafka, Datadog, Firebase, Redis, Vercel, Git, CI/CD, Jest, Playwright, Claude Code, Cursor, Codex

Interests: Quantum Computing, AI, Robotics, Cooking, Film Photography, Running, Specialty Coffee